Refining Influence Diagram For Stock Portfolio Selection

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Abstract

In this paper, we propose use of the influence diagram for stock portfolio selection. We use an algorithm that applies the mutual information as the metric to guide the refinement of the influence diagram. We applied the algorithm to the conceptual refinement of the influence diagram. We tested our algorithm to a specific domain – portfolio selection; the result is impressive compared to the S&P500. Experiments comparing our model with the other machine learning techniques are also provided.

Introduction

Previous work in the portfolio management domain has focused on the portfolio selection process- the stock picking. Portfolio managers are generally stock pickers and only occasionally market timers. Their goal is to choose between outperforming and under-performing stocks at a comparable risk level. When choosing among companies, the portfolio managers attribute much importance to the company's financial data. This data demonstrated the ability of a company to manage and profit through the economical and sociological impact of the uncertain financial world. Although this process has been applied to expert systems such as rule-based system, such system cannot adapt to the fast changing financial environment. A neural network system is another tool for stock selection process. The disadvantage of a neural network is that it requires a lot of time and data to train, which is usually very computational intensive and time consuming.

An influence diagram on the other hand provides a very good tool to model the stock picking process by using the probability to represent the domain knowledge. In Binder et al. [1], their study shows that the probabilistic network outperforms the neural network due to the prior knowledge embedded in the network. In our model, the probabilities are learned from the historic financial data and the structure of our network was built by consulting with the financial expert on portfolio selection. In order to improve the influence model there are three types of refinement that we can perform on it: (1) quantitative, (2) conceptual and (3) structure [2]. Our paper focuses on the method of the conceptual refinement by using the mutual information [3] and utility of refinement as our criteria to improve our model. Using our method we can show in our experiment that our stock portfolio selection system based on the influence diagram improves its performance with the refinement. We also factor in the human risk attitude and real time news influence on the portfolio selection system that dynamically reflects of the real world situation and extends the ability of our system.

Influence Diagrams

An influence diagram can be viewed as a special type of Bayesian network, where the value of each decision variable is not determined probabilistically, but rather is computed to meet some optimization objective.

Influence diagrams are directed acyclic graphs with three types of nodes – chance nodes, decision nodes and utility nodes. Chance nodes are usually shown as oval, represent random variables in the environment. The decision nodes usually shown as square represent the choices available to the decision-maker. Utility nodes usually shown as diamond or flattened hexagon shape, represents the usefulness of the consequences of the decisions measured on a numerical scale called the utility scale. The arcs in the graph have different meanings based on their destinations. Dependency arcs are the arcs that point to utility or chance nodes representing probability or functional dependence. For example, in figure 1 the value of the return on equity node is depend on the value of the total asset turnover node and the value of the pretax profit margin node. Informational arcs are the arcs that point to the decision nodes implying that the pointing nodes will be known to the decision-maker before the decision is made.

There are several methods for determining the optimal decision policy from an influence diagram. The first is by Howard and Matheson (1981)[5]. Their method consists of converting the influence diagram to a decision tree and solving for the optimal policy within the tree, using the exp-max labeling process. The disadvantage of this method is the enormous space required to store the tree. The second method is by Shachter (1986)[6]. His method consists of eliminating nodes from the diagram through a series of value preserving transformations. Each transformation leaves the utility intact, and at every step, the modified graph is still an influence diagram. Shachter proved that the transformation does not affect the optimal decision policy and the value of the optimal policy. However, it still requires a large amount of space to support the transformation steps. Pearl (1984)[4] has suggested a hybrid method uses branch and bound techniques to prune the search space of the converted decision tree from the influence diagram. The disadvantage of this method is the trading off time for space, so it will requires more time to obtain the optimal policy.

Portfolio Selection Model

Any model is inevitably a simplification of the real world problem. The essential issue is to decide which variables and relationships are important and which can be omitted or are redundant. The structure of the diagram was constructed by consulting with the domain expert. After several interviews with the domain expert, we came up with the portfolio selection model shown in figure 1.

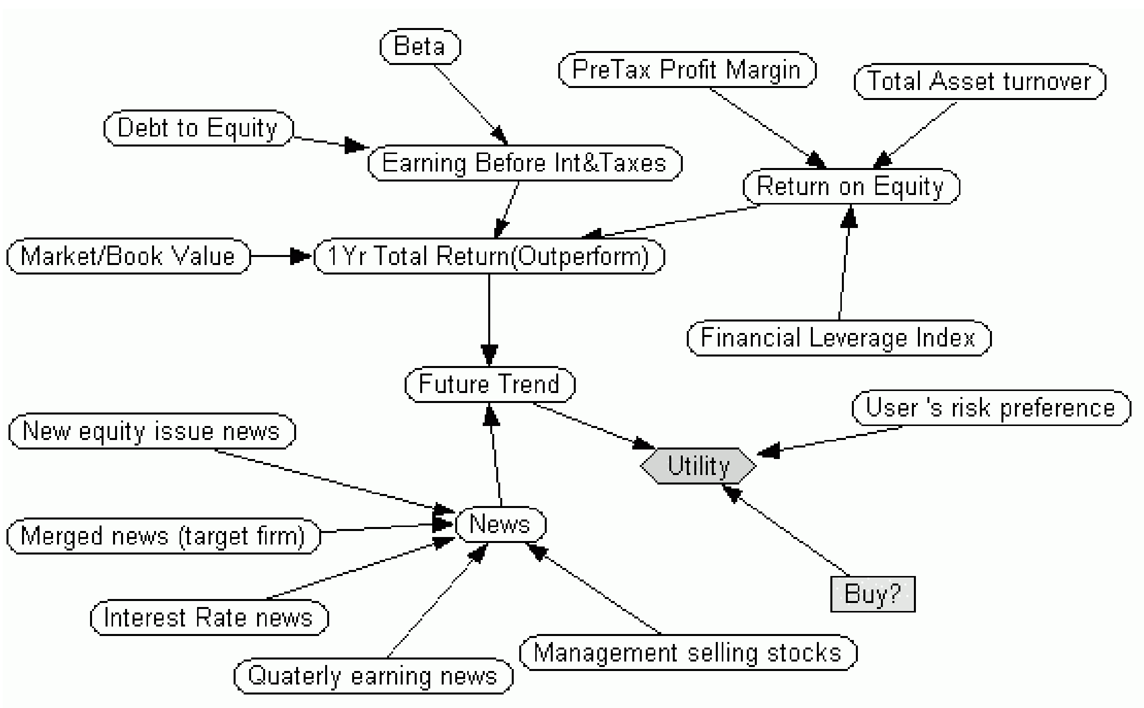

Figure 1. Influence diagram for stock portfolio selection.

Once the initial diagram was constructed, the next stage was to define the number of values for each variable. Most of the variables are continuous, but all were modeled as discrete. For the first prototype, we modeled the diagram as simple as possible, and each value of each variable was carefully defined. For example, the beta value was defined as two values each representing two ranges of beta values. Such explicit definitions are necessary to avoid ambiguity when assessing conditional probabilities.

Risk tolerance and News factors

Different portfolio managers have different preferences, under the same situation some might decide to purchase the stocks while others might decide to hold back. As we mentioned before, the goal for the portfolio manager is to maintain a profitable portfolio under a certain risk level. Therefore, risk is an important factor and varies among different portfolio managers.

Another important factor in the financial domain is the news about a certain company's situation or economical situation. This has been an intensive research area in the financial community. News has a major influence on the market trend especially in the dynamic and uncertain financial market. Timely news is an important factor for the portfolio managers. To make our model more flexible to the portfolio managers we modeled the risk attitude of the portfolio manager and the impact of the news factor into our system.

In order to model the human users risk preference and the news impact factor, we used a risk preference node to represent the risk preference of the user and a news node to represent the news factor. The news factor node has several child nodes representing different types of the news; we selected five of them by consulting with the domain expert and the research done on the event study. The risk preference node has an arc going into the utility nodes that makes the utility node risk dependent. Our goal is to use the dependency between the nodes to model the human user's risk preference and the impact of the news factor when choosing a stock. The utility U(a, r, f) represents that the utility depends on the action a (Pick action), r (Risk preference) and f (Future Trend). The expert uses subjective values representing the situations and assigns the values to the utility node.

Mutual information and Value of refinement

Mutual information is one of the most commonly used measures for ranking information sources. Here we apply this to our nodes in the network in order to provide guidance for refinement. Mutual information is based on the assumption that the uncertainty regarding any variable X represented by a probability distribution P(x) can be represented by the entropy function.

$$H(X) = -\sum_{x} P(x)\log P(x) \qquad -(1)$$

Assume that the target hypothesis is H and we want to know the uncertainty of H given X is instantiated to x, can be written

$$H(H|x) = -\sum_{h} P(h\,|\,x)\log P(h\,|\,x) \qquad -(2)$$

summing over all the possible outcome of x, we got

$$H(H|X) = -\sum_{x}\sum_{h} P(h,x)\log P(h\,|\,x) \qquad -(3)$$

where x, h are the possible values of the variable X and hypothesis H.

Then when we subtract H(H|X) from the original uncertainty in H prior to consulting X H(T), we can obtain the uncertainty reduction of H given X. This reduction is called Shannon's mutual information.

$$I(H|X) = H(H) - H(H|X)$$

$$= -\sum_{x}\sum_{h} P(h,x)\log\frac{P(h,x)}{P(h)P(x)} \qquad -(4)$$

The value of refinement is defined as the difference between the model performance before and after the refinement. The performance of the model is measure by the average expected utility of the model run on the test cases. The current model performance MP(C) is represented as:

$$MP(C) = avg_n \sum_{n} EU(a \mid x_n, C) \qquad -(5)$$

$$EU(a|x_n,C) = \max_{A} \sum_{i} U(Resulti(A))P(Resulti(A) \mid xn, C, Do(A)) \qquad -(6)$$

where x_n is the test case input to the model.

The performance of the model after the refinement MP(R) is represented as:

$$MP(R) = avg_n \sum_{n} EU(a \mid x_n, R) \qquad -(7)$$

Then the value of the refinement VR is:

$$VR = MP(R) - MP(C) \qquad -(8)$$

In the next section, we will present an algorithm that uses the two definitions we present in this section to refine the model.

Mutual Information guided refinement algorithm

In order to refine the model, we created an algorithm to provide the criteria for the knowledge engineer. The algorithm will utilize the formulae (5), (7) and (8) in the previous section. We calculated the value of refinement of the model by comparing the refined model's performance with the performance before the refinement. The refinement is guided by the mutual information, which we calculated for each node in our model based on the target hypothesis. In other words, the mutual information value represents the uncertainty reduction of the target hypothesis given the node. Here is our algorithm:

Procedure Mutual Information Guided Refinement Algorithm

```
    For all nodes except the target node
    {
        Calculates the mutual information value based on the target hypothesis.
    }
    Ranking the nodes based on the mutual information.
    Calculates the current model's performance MP(C)
    Do
    {
    Refines the highest ranking node by increasing the values in the node
    Calculates the refined model's performance MP(R)
    Calculates the VR = MP(R) – MP(C)
    If (VR < 0) then
    {
        Refines the node even further and calculates the MP(R')
        If (MP(R') > MP(R) ) and (MP(R') < MP(C))
            Refines further on the node and calculates the MP(R")
            If (MP(R") < MP(C)
```

Break and return to unrefined state for the node (Stop with this node goes to next node)

Else

Use the current refined model for the next step

MP(C) = MP(R)

Else

Goto the next node

}

else

{

Refines the node even further and calculates the MP(R')

while MP(R')-MP(C) > 0

{

MP(C) = MP(R')

refines even further

}

}

continue with the next highest ranking node

} until going through all nodes

Our algorithm calculates the mutual information value of each node based on the target hypothesis node. Moreover, ranking it according to the value, our algorithm then takes those ranked nodes and puts them under the refinement process. Our algorithm also includes a heuristic rule that when the value of refinement is less than zero, means that after the refinement (represented by r1), the model actually performs worse than before the refinement (represented by c). We then refined the node even further to see if that helps the performance of the model. If the second refinement step r2 is better than the r1, but less than c we continue another refinement step r3 on this node to see if r3 is better than c. If the r3 is not better than c, the algorithm stops the heuristic rule and uses c as the model for the next refinement step. This heuristic rule can help the model to escape from the local maximum problem, thus help improve the performance of the model.

We will see the implementation of our algorithm in the next section.

Experiments

Our model uses the S&P 500 index companies as our target. We trained our network on the 1987 to 1996 financial data we obtained from the CompuStat and tested it on 1997 data. Our performance metric is the average expected utilities from the companies that our model selected. The network structure is shown in figure 1. We applied the refinement algorithm to only the financial factor change nodes (beta, ROE, etc.), the risk tolerance and the news factor nodes remain unchanged for this experiment. The target node is the outperform node in our network. The nodes that are refined in our network are shown in table 1. Here are the tables from our experiment results.

Market/Book value	Return on Equity
Earn before tax and interest	Pretax profit margin
Total asset turnover	Debt to Equity
Financial leverage index	Beta

Table 1. The nodes that apply to the refinement algorithm.

Nodes	Number of values before the refinements	Number of values after the refinements
Market/Book value	2	7
Return on Equity	2	2
Earn before tax and interest	2	2
Pretax profit margin	2	6
Total asset turnover	2	35
Debt to Equity	2	2
Financial leverage index	2	2
Beta	2	2

Table 2. The number of values in the nodes before and after the refinement.

The average utilities of our model increase from the refinement are shown below.

Average utility before refinement	Average utility after refinement
10.067877	13.556536

Table 3. Average utility changes from the refinement process.

Figure 2. Model performance increases with the refinement steps.

Our experiment's result is encouraging. The performance of our model increases with our refinement algorithm as shown in figure 2. We tested our model on the 1997 S&P500 data and our model selected 156 out of 500 companies in the S&P 500 group. When we calculated the one-year total return on the selected companies verses the S&P 500, the portfolio selected by our model outperforms the S&P 500 in 1997 (table 4).

	S&P 500	Our portfolio
1997 Average one year total return	24.21744	38.26154

Table 4. Our portfolio verse S&P 500.

On the 1997 test data, our portfolio obtains its maximum one-year total return performance when the portfolio contains only the top 108 ranking (expected utility) companies out of the 156 selected. The top 108 companies produce an average one-year total return of 42.8264, almost twice the average of the S&P500's return.

Related Work and Comparison

The value of modeling was first addressed by Watson and Brown (1987) and Nickerson and Boyd (1980). Chang and Fung (1990) have considered the problem of dynamically refining and coarsening of state variables in Bayesian networks. However, the value and cost of performing the operations were not addressed. Control of reasoning and rational decision making under resource constraints, using analyses of the expected value of computation and considering decisions about use of alternative strategies and allocations of effort, has been explored by Horvitz(1987, 1990) and Russell and Wefald(1989). Poh and Horvitz (1993) explored the concept of expected value of refinement and applied them to structural, conceptual and quantitative refinements. Their work concentrated on providing a computational method for the criteria to perform the refinement. However, their work did not address the need of a guided algorithm to perform the node refinement throughout the network.

We also compared our work with other machine learning techniques, such as C5.0 and back-propagation neural network. We trained the C5.0 and neural network with the same training data and tested them with the same testing data we applied to our model. The neural network failed to converge due to the large variation of the training data. The C5.0, the successor of the C4.5, did produce some interesting results. The C5.0 selected 218 out of the 500 companies and the average one-year total return for that portfolio is 38.90556. Although the C5.0 slightly outperformed our model given the data, our model can handle the real time events such as financial news and the user's risk preference that the C5.0 can't. The reason for our slightly worse performance might be due to our network structure. We constructed our model based on the expert's opinion, which could be subject to expert's bias. Our algorithm currently does not address this issue on refining the structure; this will be our further improvement area to our algorithm.

Future work and Conclusion

Our mutual information guided influence diagram refinement algorithm provides a way to refine the influence diagram model thus increase the model's performance. Our experiment had shown that our algorithm improved the stock portfolio selection model and created a portfolio that outperformed the S&P 500. Our algorithm currently only concentrates on conceptual refinement; we would like to expand our algorithm in the future to quantitative and structure refinement of the influence diagram.

References

[1] J. Binder, D. Koller, S. Russell, K. Kanazawa, `` Adaptive Probabilistic Networks with Hidden Variables." *Machine Learning*, **29**, 1997, pp. 213-244.

[2] K.L. Poh and E. Horvitz, "Reasoning about the Value of Decision Model Refinement: Methods and Application.", *Proceedings of Ninth Conference on Uncertainty in Artificial Intelligence*, 1993, Morgan Kaufmann: San Francisco, CA., pp. 174-182.

[3] C Shannon and W. Weaver, *The mathematical theory of communication*, University of Illinois Press, 1949.

[4] J. Pearl, Probabilistic Reasoning in Intelligence Systems: Networks of Plausible Inference Revised Second Printing, Morgan Kaufmann: San Francisco, CA., 1997.

[5] R. A. Howard and J. E. Matheson, "Influence Diagrams", *The Principles and Applications of Decision Analysis: Vol. II*, Strategic Decisions Group, Menlo Park, CA. 1984.

[6] R. D. Shachter, "Evaluating Influence Diagram", *Operation Research*, 1987, pp. 871-872.

[7] E. J. Horvitz, "Reasoning about beliefs and actions under computational resource constraints." *Proceedings of the Third Workshop on Uncertainty in Artificial Intelligence*, 1987, pp. 429-439.

[8] S. J. Russell and P. Norvig, *Artificial Intelligence*, A Modern Approach, Prentice Hall, EngleWood Cliffs, New Jersey, 1995.

[9] S. R. Watson and R. V. Brown, "The valuation of decision analysis." *Journal of the Royal Statistical Society, 141, Part I*, 1978, pp. 69-78.

[10] R. C. Nickerson and D. W. Boyd, "The use and value of models in decision analysis." *Operation Research, 28*, 1980, pp. 139-155.

[11] K. C. Chang and R. Fung, "Refinement and coarsening of bayesian networks." *Proceedings of the Sixth Conference on Uncertainty in Artificial Intelligence*, 1990, pp. 475-482.